Code No: 117AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) What are the various components of AI program? Explain.
b) What are various problem characteristics? Explain. [7+8]

2.a) Demonstrate with an example the working of iterative deepening search algorithm.
b) What is a constraint satisfaction problem? Explain with an example. [7+8]

3.a) What is an axiomatic system and what are the properties of an axiomatic system?
b) What are the different approaches to knowledge representation? Explain. [7+8]

4.a) Explain semantic network using an example.
b) How do you make inferences using refutation in propositional logic? Explain with an example. [7+8]

5.a) Give the steps in the construction of an expert system.
b) What is a truth maintenance system and what are the components of a truth maintenance system? [7+8]

6.a) What do you mean by certainty factor? How do you calculate certainty factor?
b) What is an expert system shell? Explain. [7+8]

7.a) Give an overview of support vector machines.
b) What is a recurrent network? Explain. [7+8]

8.a) What are the steps in natural language processing? Explain.
b) What is a semantic web? How does it aid in knowledge representation? [7+8]

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